Interaction of Metric Perturbations With an Axon Tract: Demonstration of the Strong Wave Curves

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Abstract

In this note, we interact strong gravitational waves in a mathematical model which contains ephaptically coupled axons.

This paper, based on references [1, 2, 3, 4, 5], consists entirely of figures. The figures demonstrate the inclination- (geometry) and frequency- (of the wave) dependent flow of action potentials in three axons that are coupled ephaptically, numbered increasingly from top to bottom in three sub-plots. The x-axes are time in seconds and the y-axes are voltage in Volts. The frequency of the waves ranges from: 0.1 to 10 Hz, with 8 intermediate points. The angular inclinations theta range from 2 to 25 degrees, with 8 intermediate points as well. The gravitational wave strain is $h = 0.8$. The configurations (only seven are shown) determine the parameters as per the algorithm that is given here (Algorithm 1). A configuration-dependent distortion in the action potential profiles is notably present in many curves.

Algorithm 1 Parameters of the Configurations.

```
count1 = 0;
count2 = 0;
for frequency in linspace(0.1,10,10)
count1 = count1 + 1;
for theta in linspace(2,25,10)
count2 = count2 + 1;
configuration=count1*count2;
end
end
```

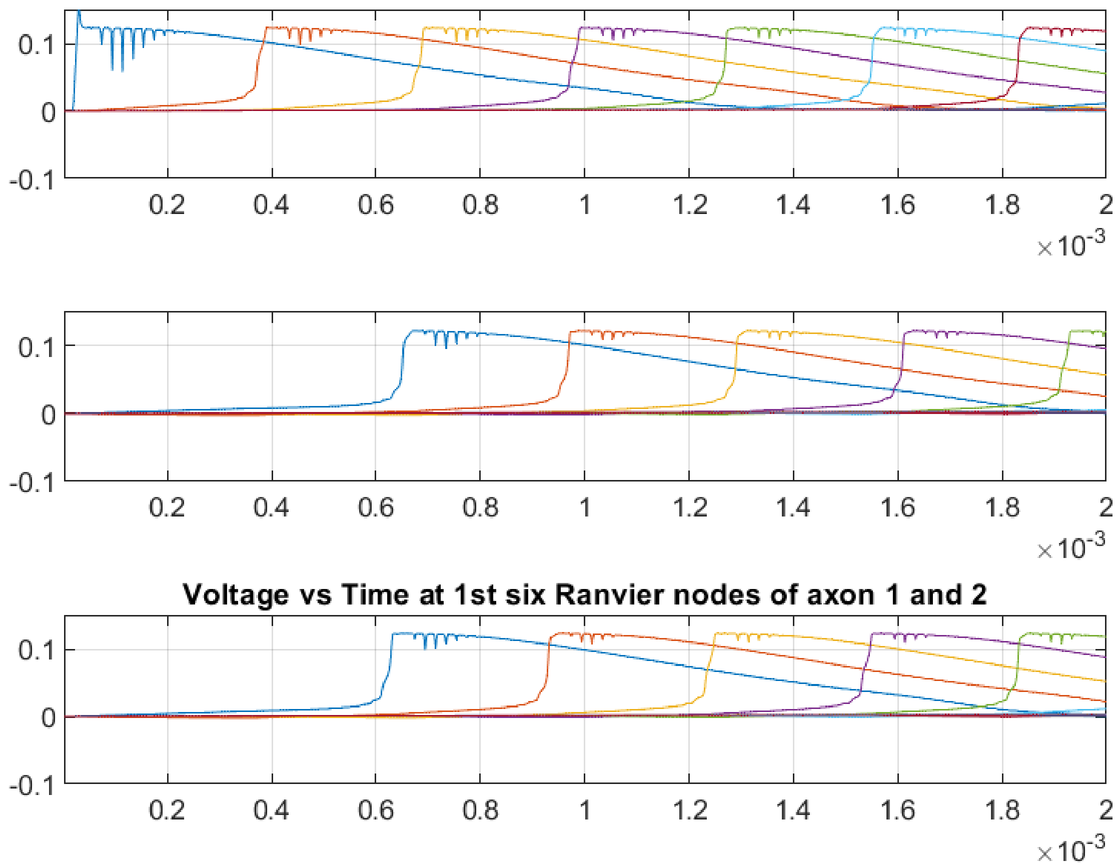

Figure 1: Configuration 1

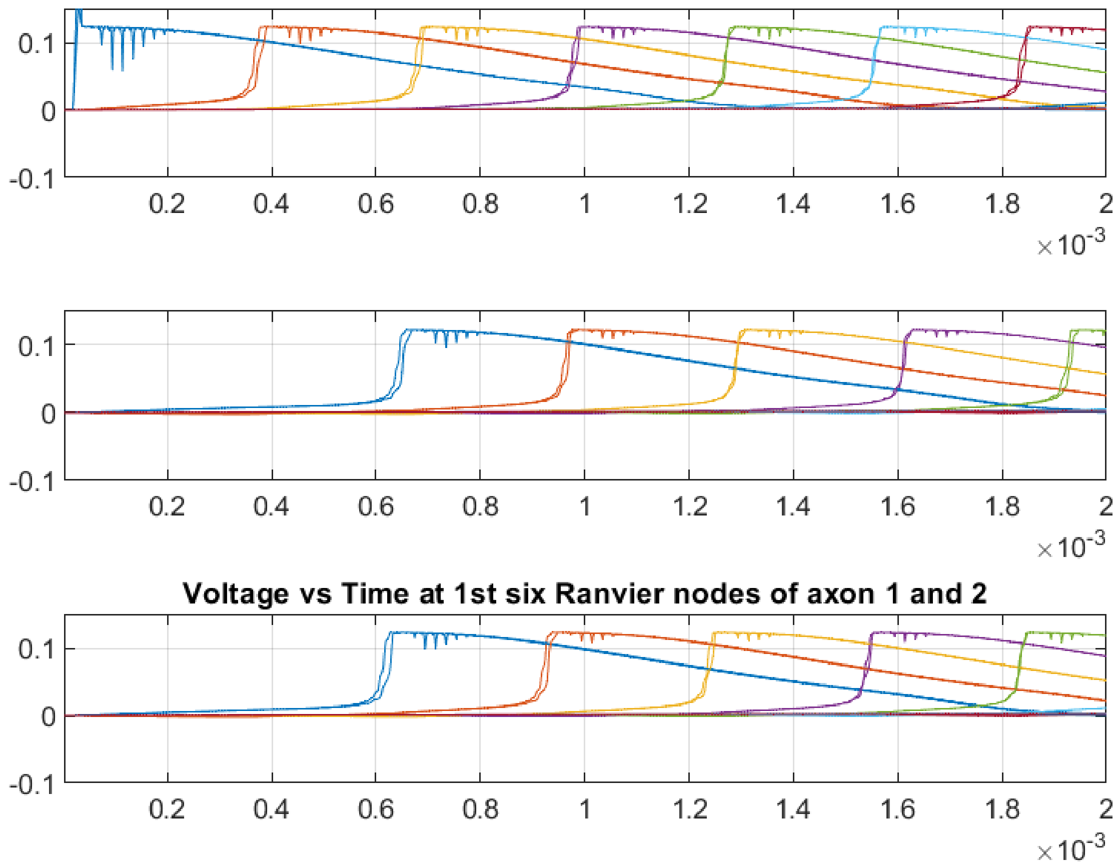

Figure 2: Configuration 2

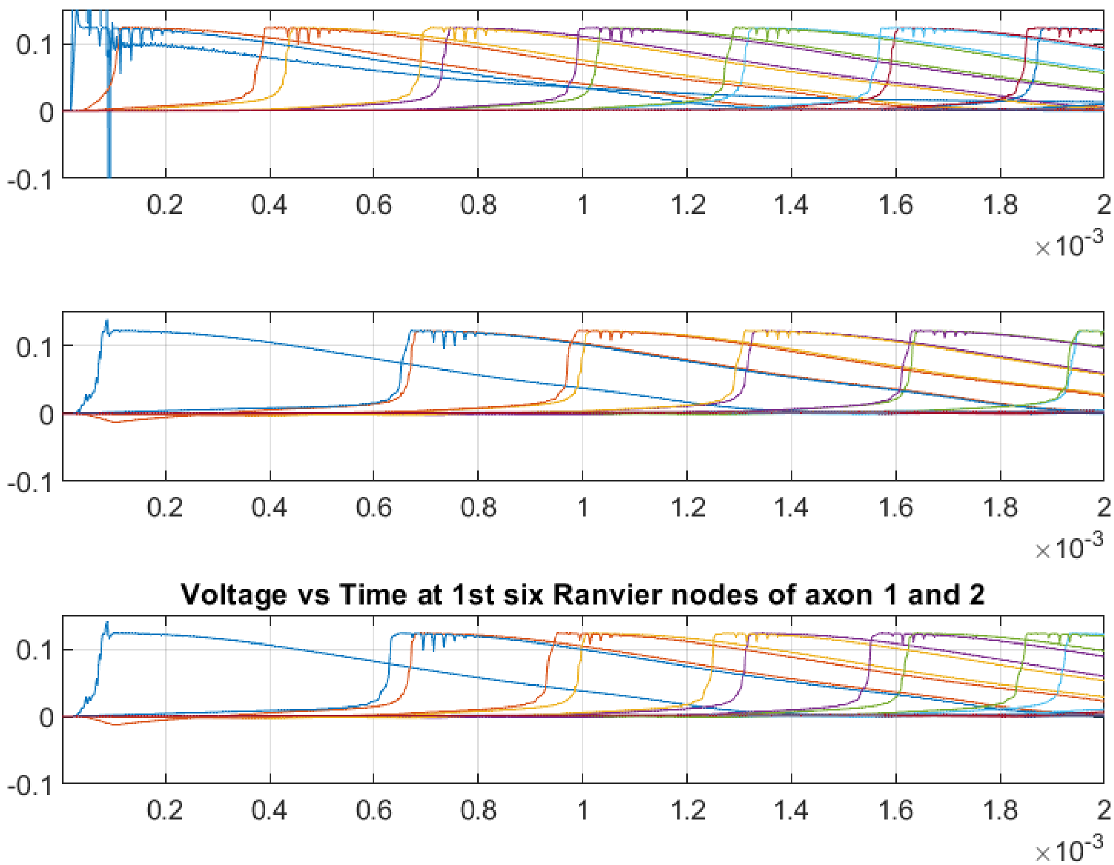

Figure 3: Configuration 3

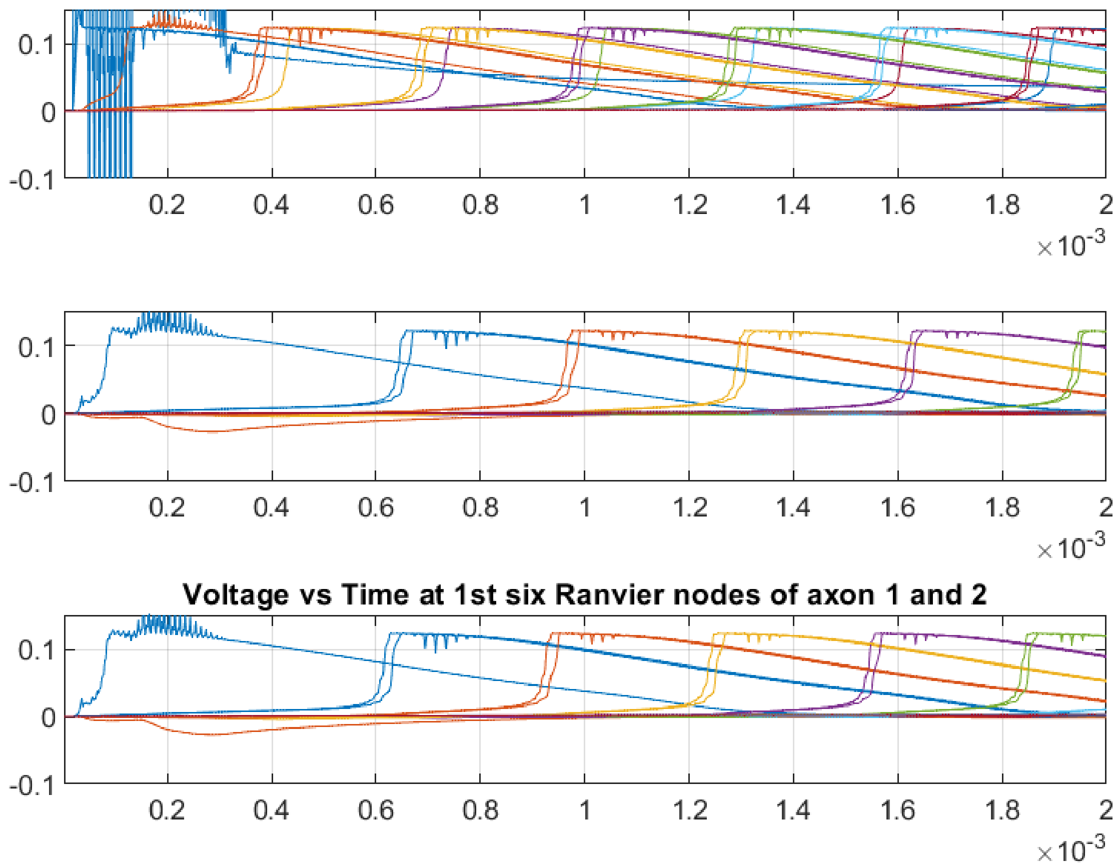

Figure 4: Configuration 4

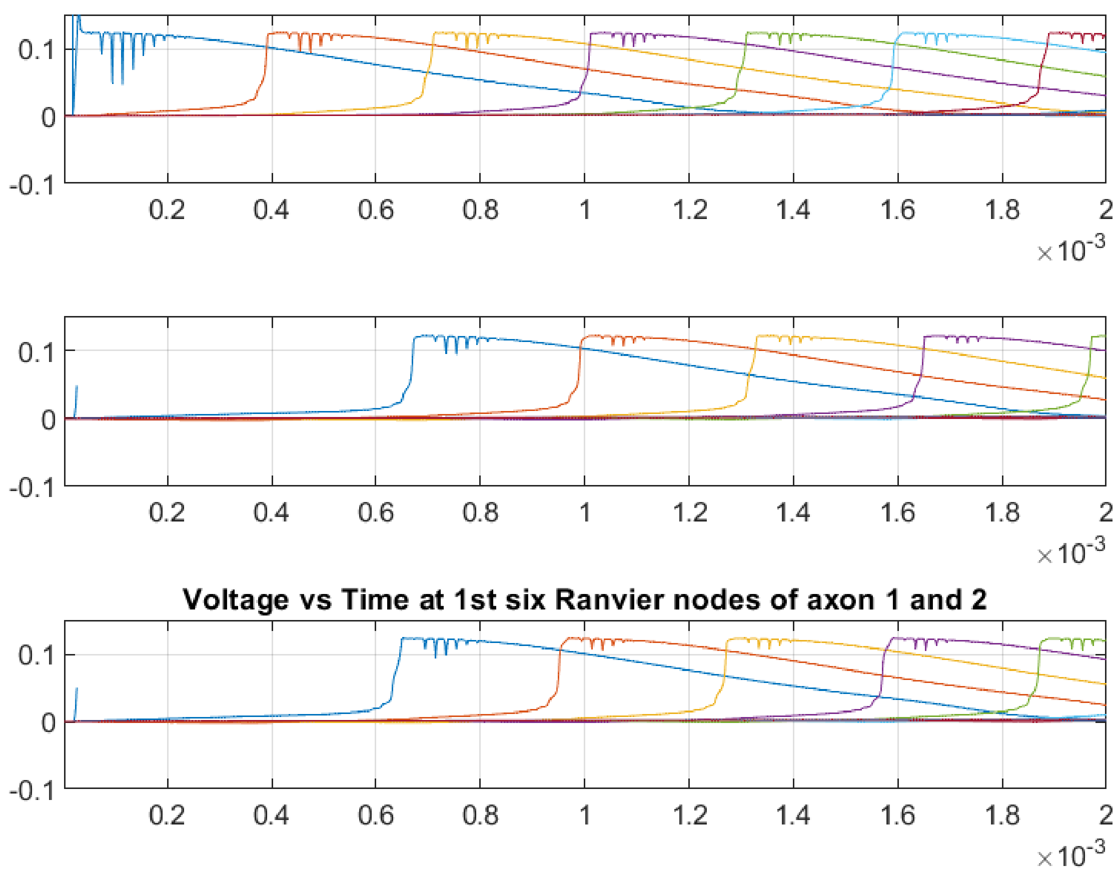

Figure 5: Configuration 5

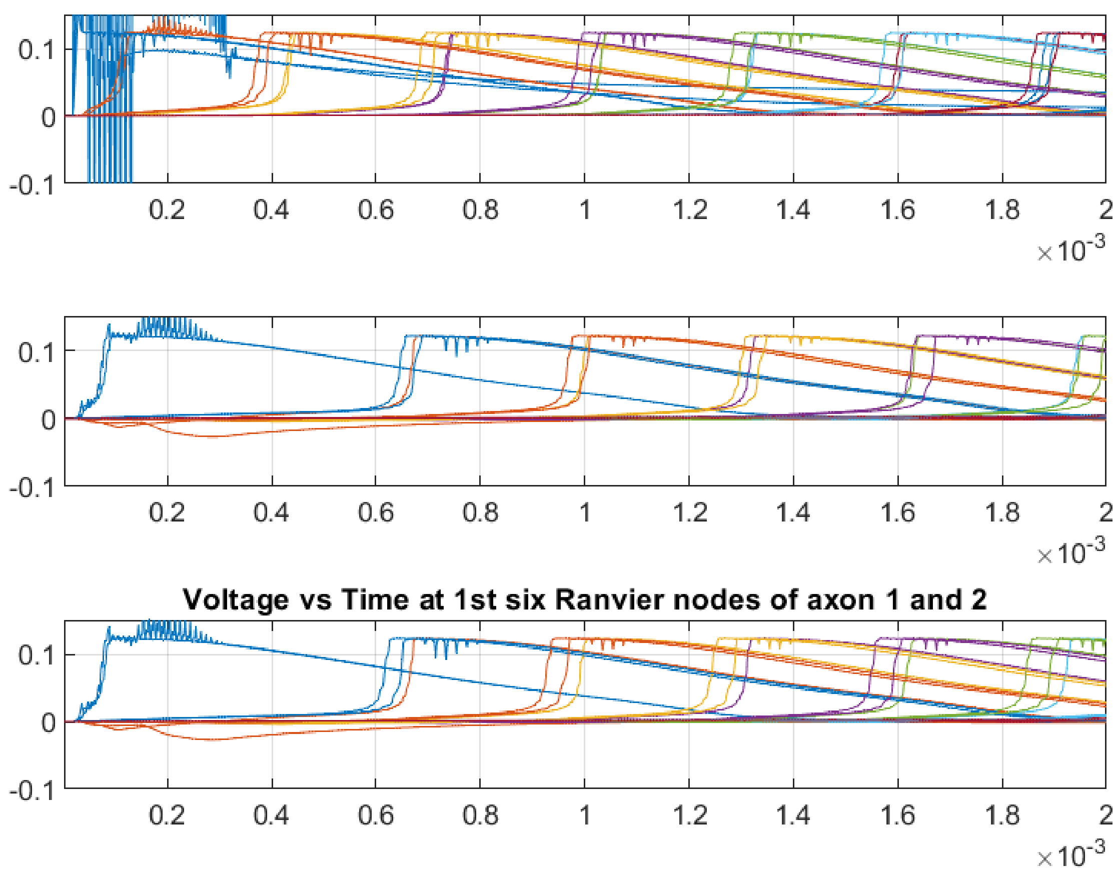

Figure 6: Configuration 6

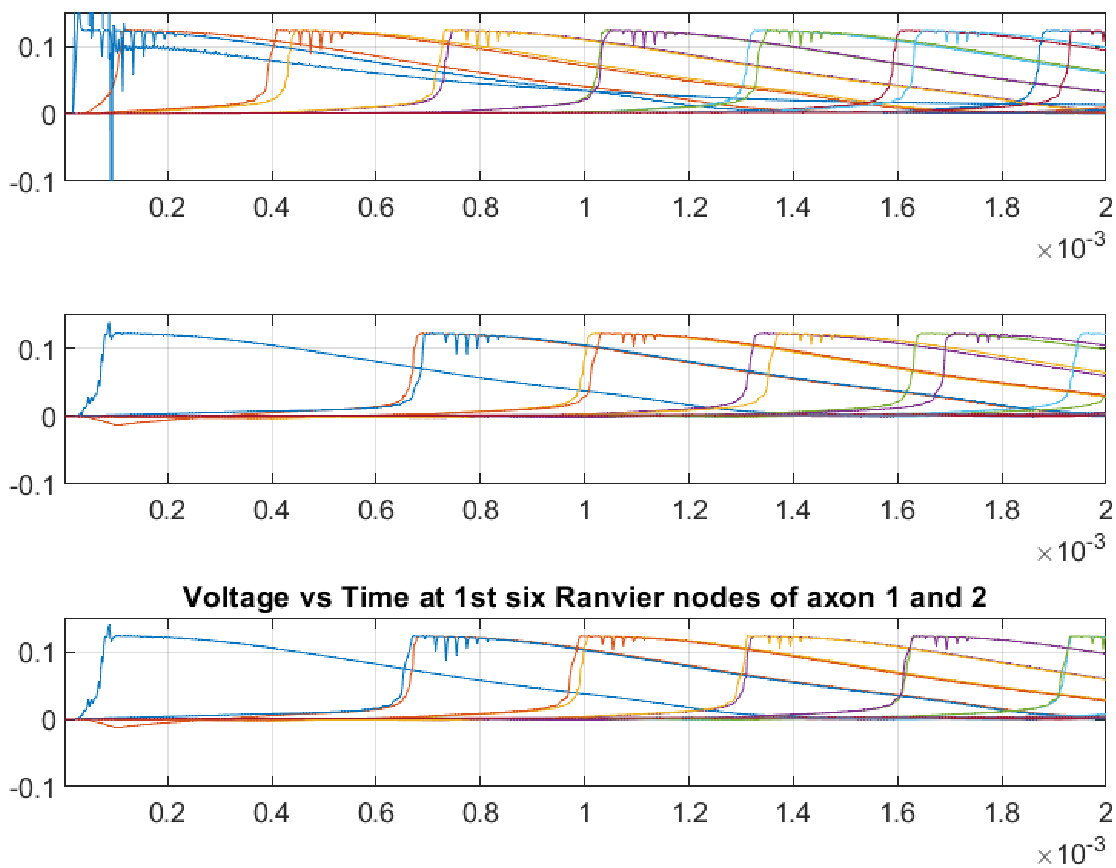

Figure 7: Configuration 7

References

[1] Aman Chawla. *On Axon-Axon Interaction via Currents and Fields*. PhD thesis, University of South Florida, 2017.

[2] Aman Chawla, Salvatore Morgera, and Arthur Snider. On axon interaction and its role in neurological networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2019.

[3] Aman Chawla, Salvatore D. Morgera, and Arthur D. Snider. Fields, geometry, and their impact on axon interaction. *Journal of Applied Mathematics and Physics*, 9(4):751–778, 2021.

[4] Sergiy Reutskiy, Enrico Rossoni, and Brunello Tirozzi. Conduction in bundles of demyelinated nerve fibers: computer simulation. *Biological cybernetics*, 89(6):439–448, 2003.

[5] Aman Chawla and Salvatore Domenic Morgera. Computer study of metric perturbations impinging on coupled axon tracts. *Physics Essays*, 36(2):160–167, 2023.